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Numerical analysis of gravity-induced coupling dynamics of keyhole and molten pool in laser welding

Chao Ma, Yue Li, Lihong Cheng, Yanqiu Zhao, Jianfeng Wang, Xiaohong Zhan *

College of Materials Science and Technology, Nanjing University of Aeronautics and Astronautics, Nanjing, 211106, China

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ABSTRACT

The dynamics and flow behavior of the molten pool and keyhole during laser welding under the influence of lateral gravity engender a distinct propensity for fluctuation, thereby manifesting a substantial influence on the quality of the laser welding process. Nevertheless, a comprehensive elucidation of these intricately interconnected dynamic mechanisms across varying gravity angles still stands as an unmet imperative. Based on the experimentally validated heat source model, the flow behavior of distorted molten pools and the dynamic evolution of keyholes under varying lateral gravity conditions were investigated. This study elucidates the impact of lateral gravity angle fluctuations on the fluid dynamics proximate to the keyhole, notably accentuated during the molten pool's growth phase. The influence of gravity manifests in a conspicuous demarcation line within the molten pool adjacent to the keyhole. The upper domain of the molten pool is subject to the Marangoni effect, inducing an upward flow, whereas the lower domain undergoes downward motion due to gravitational forces. Furthermore, as the lateral gravity angle increases, there is a notable reduction in the maximum flow velocity within the molten pool.

1. Introduction

Laser welding finds extensive utilization in aerospace, rail transportation, and related domains owing to its heightened welding efficiency, minimal welding distortions, and elevated welding precision [1–5].Within these sectors, typical structural components frequently exhibit substantial dimensions, intricate spatial orientations, and structural variability [6–9].Consequently, welding seams often contend with varying angles of gravitational influence, thereby presenting formidable challenges for laser welding in the assembly of large and intricate structures [10]. Constrained by the exigencies of practical welding conditions, notable disparities in the angles of gravitational influence are manifest in the welding seams at different locations during the laser welding process. Under distinct gravitational angles, the melt pool and keyhole within the welding seam are susceptible to conspicuous collapses [11].

To tackle the aforementioned challenges, researchers have undertaken investigations into the welding process and post-weld quality of laser welding under diverse gravitational influences. Chang et al. [12] investigated laser welding outcomes for thin Ti–6Al–4V alloy plates in both flat and horizontal orientations, highlighting that flat welding exhibited superior weld bead profiles and fewer process porosities compared to horizontal welding. Zhang et al. [13] investigated the impact of the relative angle between the laser beam and gravitational force on the formation of fill defects at the bottom of the weld, ultimately identifying the optimal tilt angle. Yuan et al. [14] discerned that during the laser welding procedure, vertically oriented upward seams display notable undercut, whereas vertically oriented downward seams manifest a significantly higher porosity rate in comparison to the upward-oriented seams.

The aforementioned studies have unveiled the impact of gravitational effects on welding quality. Nevertheless, comprehending the specific impact mechanisms and underlying principles presents a formidable challenge when relying solely on experimental methods. In order to delve deeper into the behavior of the keyhole and molten pool in laser welding under the influence of gravity, a numerical simulation is imperative for elucidating the intricacies of heat and mass transfer phenomena [15,16]. In order to comprehend the intricacies of heat and mass transfer phenomena occurring within the molten pool, numerous researchers have engaged in thermal-fluid coupling simulations to attain a more comprehensive comprehension of the laser welding process. Guo et al. [17] scrutinized the quantitative ramifications of various gaps on

* Corresponding author.
E-mail address: xiaohongzhan_nuaa@126.com (X. Zhan).

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the characteristics of the molten pool and the formation of bubbles in laser welding. Their research unveiled the intricate interplay between solidification and melting processes at the bottom of molten pool. Yin et al. [18] employed a blend of experimental analysis and numerical simulations to elucidate the characteristics of pores and the underlying evolution mechanism during the laser mirror welding of 2219 aluminum alloy. Zhou et al. [19] conducted relevant studies on vertically upward laser mirror welding, establishing the variations in the influence of gravity on flow velocities across different cross-sections.

In prior research, the scrutiny of gravity's impact on the weld pool predominantly centered on scenarios where the gravity direction was non-perpendicular to both the welding direction and the welding plane, with a particular emphasis on the keyhole's rear wall. In practical production processes, there exists a situation where the gravity direction is perpendicular to the welding direction but not perpendicular to the welding plane. This scenario is referred to as the lateral gravity angle (see Fig. 1). Inadequate scholarly emphasis has been placed on the examination of the lateral gravity angle, resulting in a lack of comprehensive exploration into the flow dynamics and thermal dispersion within the weld pool under varying lateral gravity angles. Gravity acting in this lateral direction exerts a discernible impact on both the keyhole growth and stability phases, leading to pronounced fluctuations in the keyhole. Moreover, a rigorous investigation into keyhole oscillation and stress conditions in this context remains imperative. This study endeavors to elucidate the dynamic behavior of the molten pool and keyhole during laser welding under varying lateral gravity angles.

To attain this objective, a three-dimensional coupled thermal-fluid model has been meticulously formulated. In order to explore the dynamic evolution of the molten pool and keyhole during laser welding under different lateral gravity angles, precise calibration of the molten pool is performed. Extensive investigation of the dynamic behavior of the keyhole and molten pool during laser welding is conducted, based on the flow patterns observed in multiple cross-sections. This serves as a foundational insight for the utilization of laser welding in the fabrication of intricate and sizable structures composed.

2. Experimental details

The materials employed in this investigation comprise Ti–6Al–4V titanium alloy and optical glass, distinguished by the titanium alloy sheet dimensions of 100 mm × 50 mm × 2 mm and the optical glass dimensions of 100 mm × 50 mm × 2 mm. Table 1 furnishes the chemical composition of the Ti–6Al–4V titanium alloy employed in this study. Preceding the commencement of the welding process, meticulous removal of surface oxides within the welding zone was achieved through abrasive sanding, followed by thorough cleansing of the welding surface with anhydrous ethanol. Subsequently, a scrupulous drying process was executed to ensure complete desiccation and the restoration of the pristine surface, thereby preparing the workpiece for the welding procedure.

In this research, the "sandwich method" was adopted for the observation of laser welding melt pool and keyhole morphology under the influence of lateral gravitational angles. High-speed imaging systems, facilitated by optical glass, were employed to observe the laser welding melt pool and keyhole morphology under the influence of lateral gravity, utilizing different optical filter.

This study established an experimental platform to achieve stable welding at different lateral gravity angles. The schematic diagrams illustrating laser welding in the flat position, 15° position, and 30° position are depicted in Fig. 2(b), where the angle is defined as the inclination from the vertical plane of the workpiece in the direction opposite to gravity. Moreover, the welding was conducted utilizing a laser power of 950 W and maintaining a defocused distance of 0 mm, at a welding speed of 1.80 m/min. Pure argon gas was employed as the shielding gas with a flow rate of 25 L/min. Additionally, an 8 mm diameter shielding gas nozzle, positioned at a 45° inclination, was utilized to safeguard the weld.

3. Modeling

In the laser welding process of Ti–6Al–4V titanium alloy, the dynamic flow of high-temperature liquid metal and the oscillations of the keyhole within the molten pool are intricately influenced by both the laser heat source and the gravitational forces. Therefore, a comprehensive heat flow model was meticulously constructed to account for the intricate heat and mass transfer phenomena, which were subsequently analyzed through numerical simulations within the context of laser welding subjected to gravitational effects. The critical thermophysical properties of the base material, vital for the simulation, have been meticulously compiled in Table 2 and have been sourced from reputable references. [20–22].

Laser welding of titanium alloys encompasses a wide spectrum of intricate physical phenomena, including heat transfer, mass transfer, phase transition, fluid dynamics, plasma generation, and more. To streamline the computational process, not all pertinent physical factors have been incorporated into this mathematical model. The following assumptions were employed to streamline the calculations and improve the model's computational efficiency.

(1) During laser welding, the fluid is regarded as an incompressible Newtonian fluid with laminar flow.
(2) The plume and vapor dynamics caused by evaporation are ignored, but the recoil pressure is calculated.
(3) Only considering the heat transfer between the material and its surroundings, while disregarding the protective gas.
(4) The laser welding process is transient, and the initial temperature of the titanium alloy joint is 300 K.

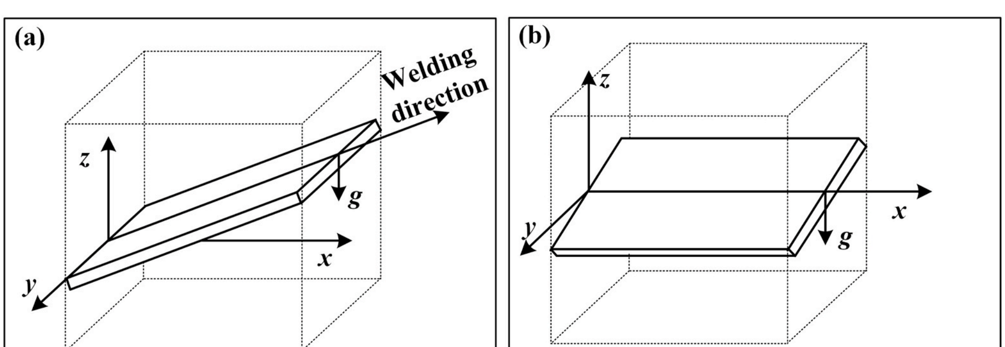

Fig. 1. Gravity angle illustration: (a) Gravity angle from the rear side; (b) Lateral gravity angle.

Table 1
Chemical compositions of Ti–6Al–4V titanium alloy.

Element	Fe	C	H	O	N	Al	V	Ti
wt%	≤0.30	≤0.10	≤0.015	≤0.20	≤0.05	5.5–6.8	3.5–4.5	Bal

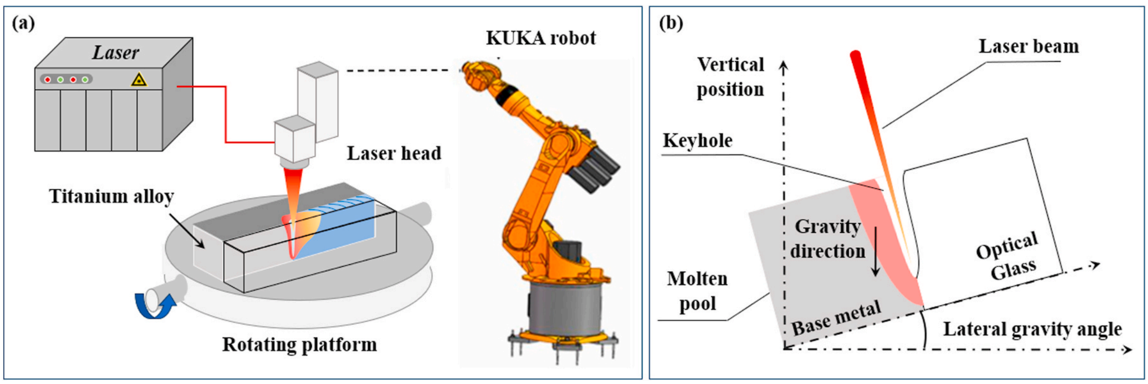

Fig. 2. Laser welding principle diagram: (a) Welding equipment diagram; (b) Diagram of laser welding at different angles of gravity.

Table 2
Thermophysical properties of Ti–6Al–4V titanium alloy.

Physical	Unit	Value
Liquidus temperature	K	1878
Solidus temperature	K	1928
Boiling point	K	3133
Liquidus density	$kg{\cdot}m^{-3}$	4000
Solidus density	$kg{\cdot}m^{-3}$	4510
Thermal conductivity of liquid	$W{\cdot}m^{-1}\ K^{-1}$	29
Thermal conductivity of solid	$W{\cdot}m^{-1}\ K^{-1}$	21
Specific heat of liquid	$J{\cdot}kg^{-1}\ K^{-1}$	730
Specific heat of solid	$J{\cdot}kg^{-1}\ K^{-1}$	670
Dynamic viscosity	$kg{\cdot}m^{-1}\ s^{-1}$	0.005
Coefficient of thermal expansion		8×10^{-6}
Thermal emissivity		0.4
Ambient temperature	K	300
Convective heat transfer coefficient	$W{\cdot}K^{-1}\ m^{-2}$	15
Latent heat of fusion	$J{\cdot}kg^{-1}$	1.12×10^{6}
Latent heat of evaporation	$J{\cdot}kg^{-1}$	1.49×10^{6}
Surface tension	$N{\cdot}m^{-1}$	1.4
Temperature gradient of surface tension	$N{\cdot}m^{-1}\ K^{-1}$	-0.26×10^{-3}

3.1. Governing equations

In the laser welding process of Ti–6Al–4V titanium alloy, the physical variables governing the flow of high-temperature liquid metal and the heat transfer behavior in the molten pool adhere to the principles of mass, momentum, and energy conservation. These principles are delineated through a set of three-dimensional governing equations, elucidated as follows.

Mass conservation equation [23]:

$$\frac{\partial \rho}{\partial t}+\frac{\partial(\rho k)}{\partial x}+\frac{\partial(\rho l)}{\partial y}+\frac{\partial(\rho n)}{\partial z}=0 \tag{1}$$

where ρ is the material density, t is the welding time and k, l and n are the fluid velocity components in the X, Y and Z directions respectively.

Momentum conservation equation [24]:

X Direction:

$$\begin{aligned}&\frac{\partial(\rho k)}{\partial t}+\frac{\partial(\rho kk)}{\partial x}-\frac{\partial(\rho k v_0)}{\partial x}+\frac{\partial(\rho kl)}{\partial y}+\frac{\partial(\rho kn)}{\partial z}\\&=-\frac{\partial P}{\partial x}+\frac{\partial}{\partial x}\left(\mu\frac{\partial k}{\partial x}\right)+\frac{\partial}{\partial y}\left(\mu\frac{\partial k}{\partial y}\right)+\frac{\partial}{\partial z}\left(\mu\frac{\partial k}{\partial z}\right)+S_k\end{aligned} \tag{2}$$

Y Direction:

$$\begin{aligned}&\frac{\partial(\rho l)}{\partial t}+\frac{\partial(\rho lk)}{\partial x}-\frac{\partial(\rho l v_0)}{\partial x}+\frac{\partial(\rho ll)}{\partial y}+\frac{\partial(\rho\ ln)}{\partial z}\\&=-\frac{\partial P}{\partial y}+\frac{\partial}{\partial x}\left(\mu\frac{\partial l}{\partial x}\right)+\frac{\partial}{\partial y}\left(\mu\frac{\partial l}{\partial y}\right)+\frac{\partial}{\partial z}\left(\mu\frac{\partial l}{\partial z}\right)+S_l\end{aligned} \tag{3}$$

Z Direction:

$$\begin{aligned}&\frac{\partial(\rho n)}{\partial t}+\frac{\partial(\rho nk)}{\partial x}-\frac{\partial(\rho n v_0)}{\partial x}+\frac{\partial(\rho nl)}{\partial y}+\frac{\partial(\rho nn)}{\partial z}\\&=-\frac{\partial P}{\partial z}+\frac{\partial}{\partial x}\left(\mu\frac{\partial n}{\partial x}\right)+\frac{\partial}{\partial y}\left(\mu\frac{\partial n}{\partial y}\right)+\frac{\partial}{\partial z}\left(\mu\frac{\partial n}{\partial z}\right)+S_n\end{aligned} \tag{4}$$

where P is the fluid pressure, v_0 is the welding speed and S_k, S_l and S_n are the momentum source terms in the X, Y and Z directions respectively.

Energy conservation equation [25]:

$$\begin{aligned}&\frac{\partial(\rho H)}{\partial t}+\frac{\partial(\rho kH)}{\partial x}-\frac{\partial(\rho v_0 H)}{\partial x}+\frac{\partial(\rho lH)}{\partial y}+\frac{\partial(\rho nH)}{\partial z}\\&=\frac{\partial}{\partial x}\left(k\frac{\partial T}{\partial x}\right)+\frac{\partial}{\partial y}\left(k\frac{\partial T}{\partial y}\right)+\frac{\partial}{\partial z}\left(k\frac{\partial T}{\partial z}\right)+S_E\end{aligned} \tag{5}$$

where k is the thermal conductivity of the material, H is the mixing enthalpies and S_E is the energy source term. For the energy source term, S_E could be written as follow [25]:

$$\begin{aligned}S_E&=Q-m_v L_v-\left[\frac{\partial}{\partial t}(\rho\Delta H)+\frac{\partial}{\partial x}(\rho u\Delta H)+\frac{\partial}{\partial y}(\rho v\Delta H)+\frac{\partial}{\partial z}(\rho w\Delta H)\right]\\&+\frac{\partial}{\partial x}(\rho v_0\Delta H)\end{aligned} \tag{6}$$

where Q is the laser heat source, m_v is the mass transfer, L_v is the latent heat of evaporation and ΔH is the latent heat of phase transition.

3.2. Computational domains and boundary conditions

The computational domain comprises of the liquid-phase region (titanium domain) and gas region (argon domain), as illustrated in Fig. 3. To incorporate symmetry into the model, the symmetric plane is utilized to simulate half of the actual weld joint within the computational domain. The lateral boundaries of the liquid-phase region are represented as immovable walls in the model. The pressure inlet is situated at both the upper and lower surfaces of the gas region, while the remaining surfaces are designated as pressure outlets. Before

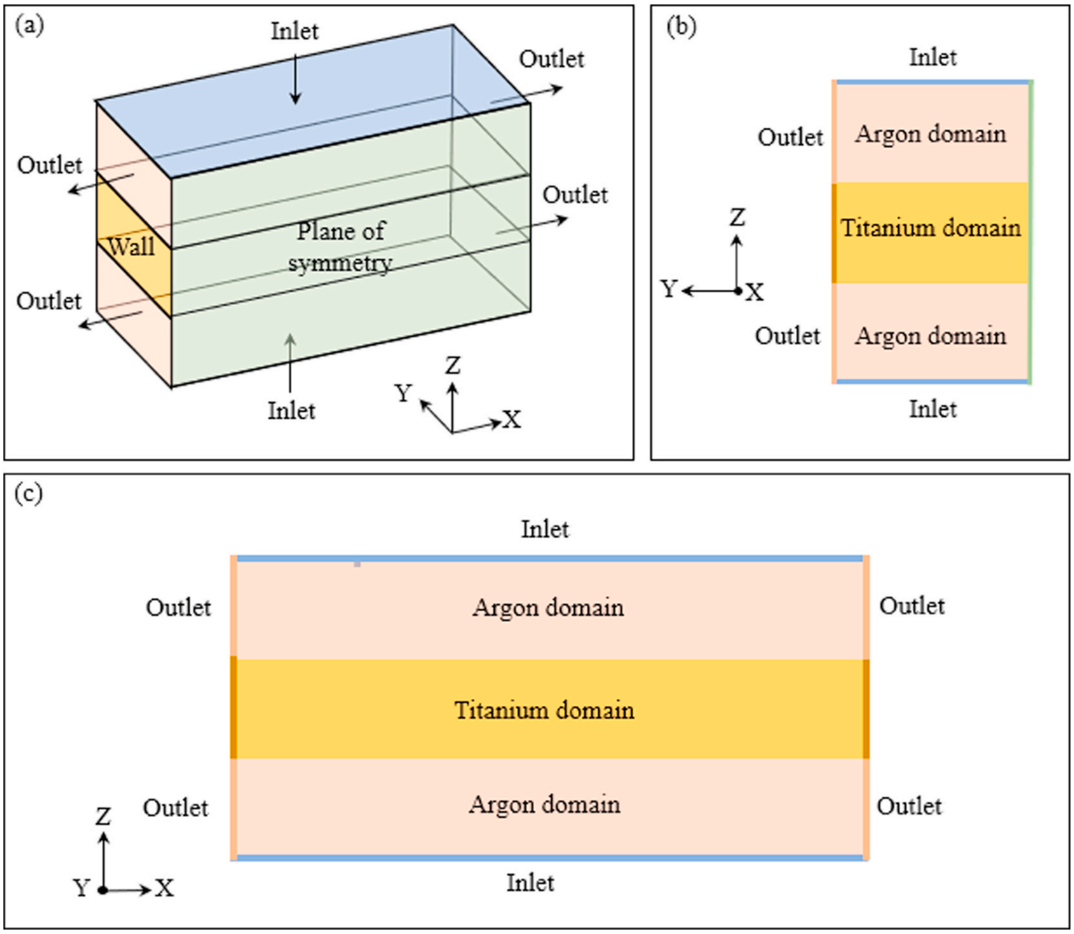

Fig. 3. Boundary conditions setup during welding: (a) Overall 3D map; (b) Y-Z plane; (c) X-Z plane.

commencing the welding process, it is essential that the argon domain is maintained at ambient temperature and standard atmospheric pressure.

In the realm of laser welding, the weldment's surface undergoes the intricate interplay of multiple thermal phenomena, encompassing laser-induced heating, convective heat transfer, radiative heat flux, and the consequential metal evaporation. The pertinent energy boundary conditions are delineated as follows [26]:

$$k\frac{\partial T}{\partial \vec{n}} = Q - q_c - q_r - q_e \tag{7}$$

where $\vec{n}$ is the laser heat source, q_c is the convective heat loss, q_r is the thermal radiation energy loss and q_e is the evaporation energy loss.

$$q_c = h_c(T - T_0) \tag{8}$$

$$q_r = \varepsilon\sigma\left(T^4 - {T_0}^4\right) \tag{9}$$

$$q_e = m_v L_v \tag{10}$$

where T is the weldment temperature, T_0 is the ambient temperature, h_c is the convective heat transfer coefficient, ε is the radiative heat transfer coefficient and σ is the Stephan-Boltzmann constant.

3.3. Laser heat source model

In the realm of simulation modeling, the adoption of a fitting heat source model has the potential to significantly enhance the precision and veracity of the simulation outcomes. Laser deep penetration welding is distinguished by a diminutive molten pool, focalized energy, and elevated temperatures within the central and keyhole sectors of the molten pool. Hence, this paper employs a composite heat source model to simulate the laser welding process. The composite heat source model comprises a Gaussian surface heat source and a Gaussian rotary body heat source, depicted in Fig. 4.

The Gaussian surface heat source model is depicted as follows [27]:

$$q_s(x,y) = \frac{\eta_s Q_s}{\pi {r_e}^2}\exp\left(-\frac{C_s(x^2+y^2)}{{r_e}^2}\right) \tag{11}$$

where η_s is the concentration coefficient of Gaussian surface heat source, Q_s is the Gaussian surface heat source effective energy, r_s is the Effective radius of heat source, and C_s is correction coefficient of Gaussian surface heat source, which continuously adjusted during the simulation process, and finally matches the experimental verification results completely.

The Gaussian rotary body heat source model is depicted as follows [28]:

$$q_r(x,y,z) = \frac{\eta_r Q_r}{\pi h {r_e}^2 (1 - e^{-3})}\exp\left(-\frac{C_r(x^2+y^2)}{{r_e}^2 \ln(h/z)}\right) \tag{12}$$

where η_r is the concentration coefficient of Gaussian rotary body heat source, Q_s is the surface heat Gaussian rotary body heat source, h is the effective depth of Gaussian rotary body heat source, and C_s is correction coefficient of Gaussian rotary body heat source.

The Combination heat source model is depicted as follows:

$$\eta Q = Q_s + Q_r \tag{13}$$

where Q is total laser power, η is the absorptivity of base metal.

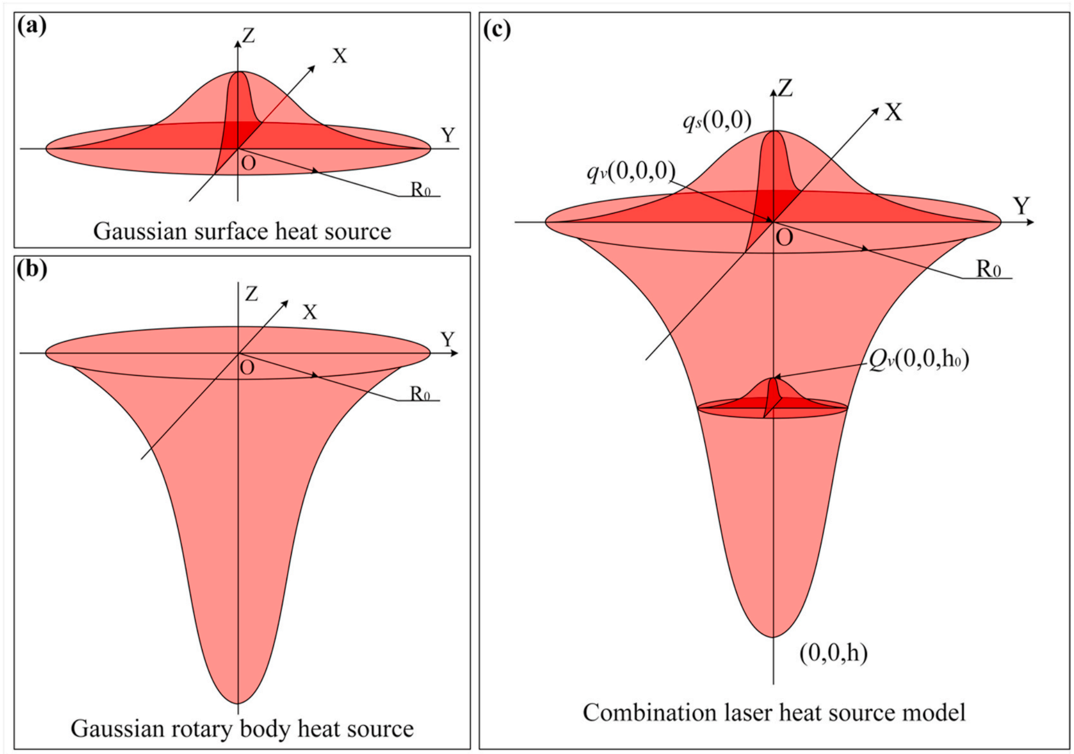

Fig. 4. Heat source model: (a) Gaussian surface heat source; (b) Gaussian rotary body heat source; (c) Combination heat source model.

4. Result and discussion

4.1. Simulation model validation

Using the parameters previously specified, the comparative results of simulation and experimentation for laser welding in the flat position are presented in Fig. 5. It is evident that the simulated weld profile closely aligns with the experimental results' profile (illustrated by the red dashed line). This signifies the feasibility and congruence of the thermal flux model employed in this study with the actual conditions.

4.2. Molten pool morphology

Fig. 6 elucidates the simulated outcomes of the surface temperature field within the laser welding molten pool at distinct time intervals and lateral gravity angles. These simulation results were obtained during both the molten pool growth stage (t = 20 ms) and the molten pool stabilization stage (t = 90 ms). The findings illustrate that the surface configuration of the molten pool exhibits similarity across varying lateral gravity angles. Measurements reveal that throughout the molten pool growth stage, the widths of the molten pool at lateral gravity angles

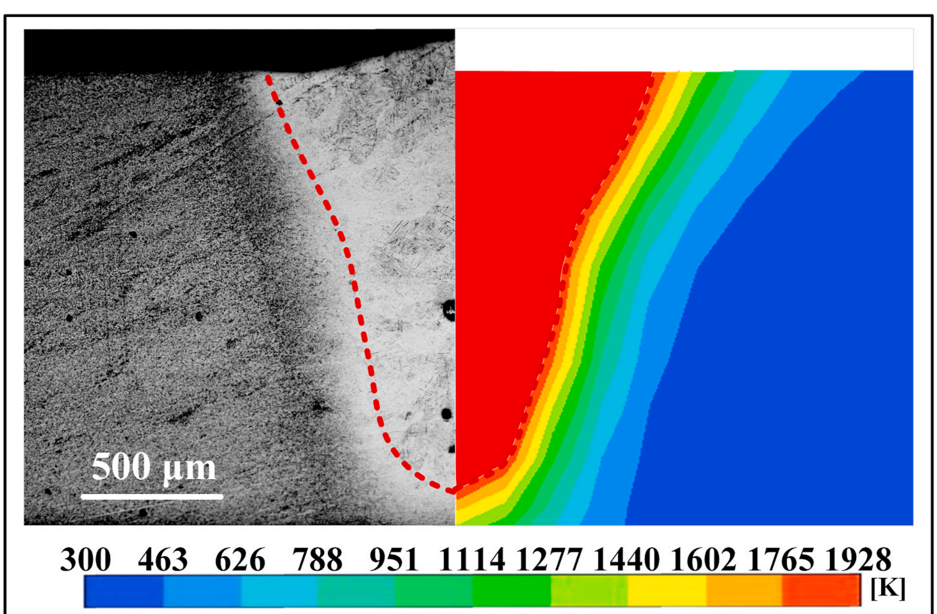

Fig. 5. The numerical and experimental weld profile in welding.

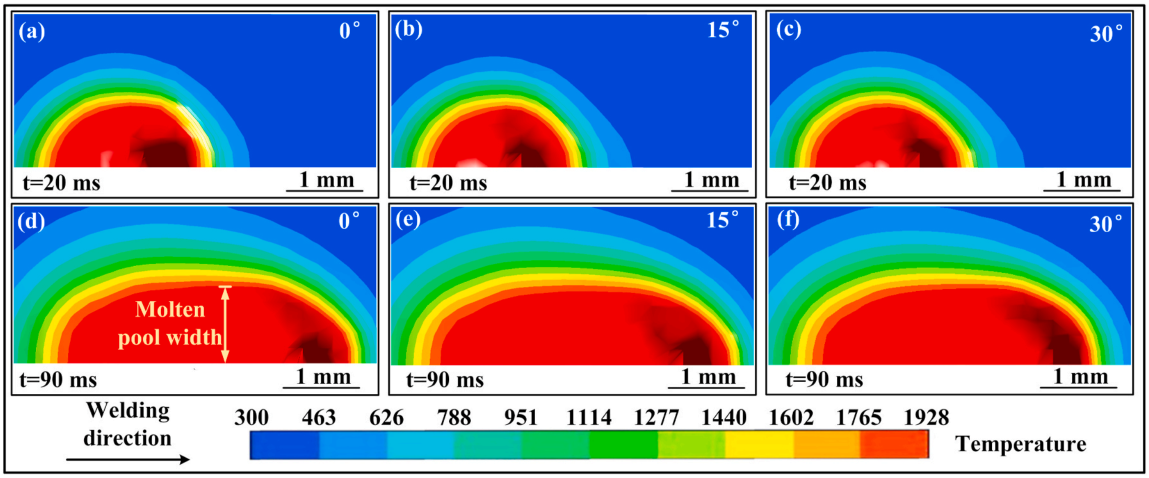

Fig. 6. Temperature field results on the upper surface and longitudinal section of the molten pool under different lateral gravity angle: (a, d) lateral gravity angle = 0°; (b, e) lateral gravity angle = 15°; (c, f) lateral gravity angle = 30°.

of 0°, 15°, and 30° amount to 0.80 mm, 0.77 mm, and 0.79 mm, respectively. During the molten pool stabilization stage, the molten pool widths measure 0.98 mm, 0.94 mm, and 0.96 mm. It is evident that varying lateral gravity angles (ranging from 0° to 30°) exert negligible influence on the configuration and dimensions of the molten pool.

Moreover, the temperature field distribution along the longitudinal cross-section of the molten pool was examined across different lateral gravity angles, as depicted in Fig. 7. The results illustrate that variations in the lateral gravity angle exert negligible influence on the contour and depth of the molten pool during its growth phase. This discovery implies that the downward motion of the molten pool is scarcely affected by alterations in the lateral gravity angle. Hence, at 20 ms, the depth of the molten pool exhibits marginal fluctuations, ranging between 1.72 mm and 1.84 mm, with the lateral gravity angle ascending from 0° to 30°. Upon achieving a stable state of the molten pool at t = 90 ms, the fluctuation in molten pool depth diminishes even further, exhibiting a range of variation between 1.78 mm and 1.82 mm. Hence, as the laser welding process advances, the impact of gravitational angle on the molten pool depth progressively diminishes, ultimately leading to an insignificantly negligible effect on the molten pool depth.

Fig. 8 depicts the simulated outcomes of the cross-sectional temperature field within the keyhole region of the laser welding pool at diverse time intervals, considering various lateral gravity angles. The findings suggest that during the pool growth stage (t = 20 ms), varying lateral gravity angles exert a notable impact on the depth of the keyhole region within the molten pool. While altering the lateral gravity angle within the range of 0°–30°, corresponding variations in the pool depth within the keyhole region are observed at 1.7969 mm, 1.6406 mm, and 1.5625 mm. The modification of the lateral gravity angle effectively mitigates the downward expansion tendency of the molten pool, with a particularly discernible effect observed within the keyhole region where this alteration is most conspicuous. With the escalation of the lateral gravity angle from 0° to 15°, there is a corresponding reduction in keyhole depth, diminishing from 1.6719 mm to 1.3906 mm. Likewise, at a lateral gravity angle of 30°, the keyhole depth diminishes to 1.0781 mm. Furthermore, it is discernible from the graph that the keyhole at 30° exhibits a more flattened profile in contrast to the keyhole at 0°, inclining towards the gravitational direction. This observation underscores the substantial impact of gravity on the keyhole formation process during the pool growth stage. The gravitational influence prompts the keyhole to incline towards the gravitational force, consequently leading to a reduction in both the keyhole and molten pool

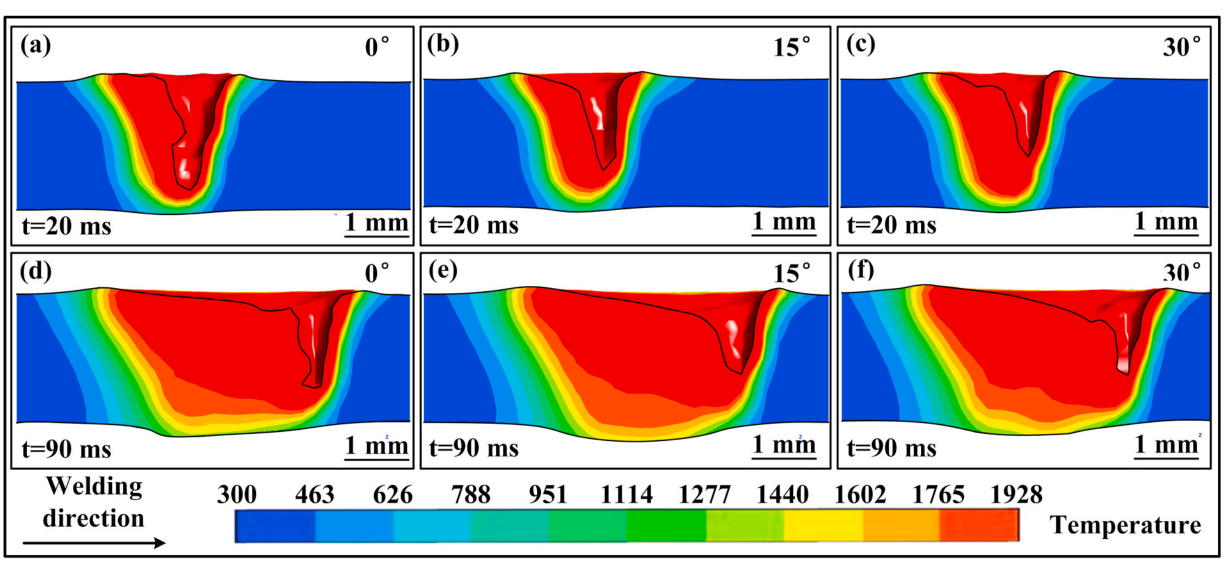

Fig. 7. Temperature field results for longitudinal sections of the molten pool under different lateral gravity angle: (a, d) lateral gravity angle = 0°; (b, e) lateral gravity angle = 15°; (c, f) lateral gravity angle = 30°.

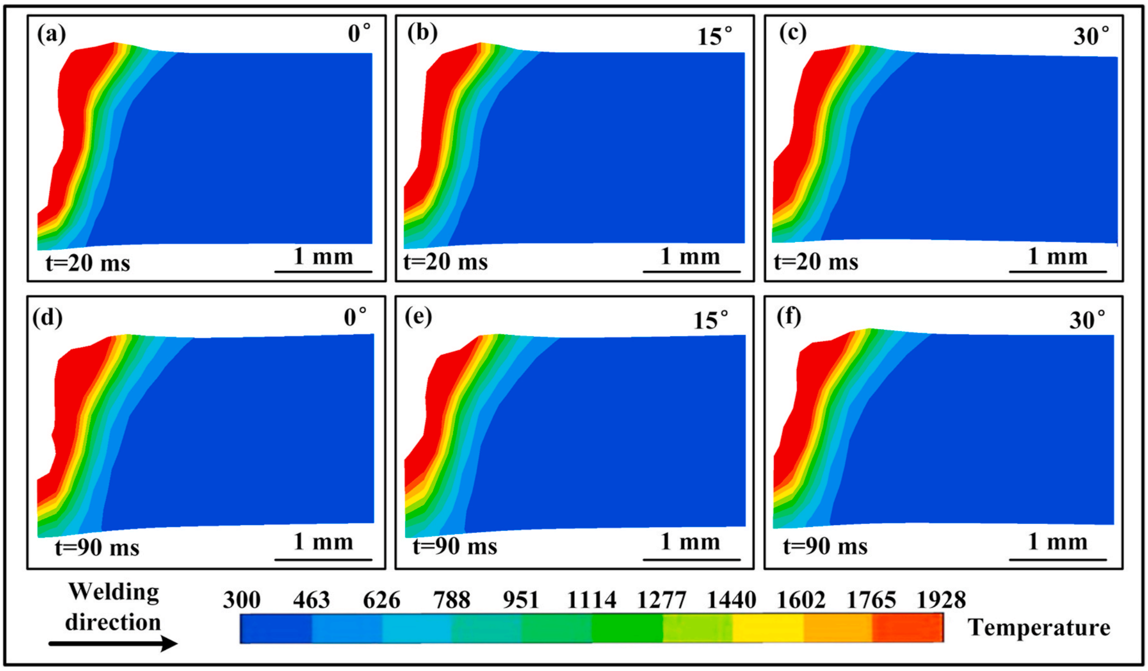

Fig. 8. Temperature field results for cross-sectional sections within the keyhole region of the molten pool under different lateral gravity angle: (a, d) lateral gravity angle = 0°; (b, e) lateral gravity angle = 15°; (c, f) lateral gravity angle = 30°.

depths.

The simulated outcomes of the temperature distribution across the cross-sectional keyhole region within the molten pool, specifically during the pool stabilization phase (t = 90 ms), are visually depicted in Fig. 8. The findings reveal that the depth of the keyhole region within the molten pool, particularly during the pool stabilization stage (t = 90 ms), is markedly responsive to variations in lateral gravity angles. While altering the lateral gravity angle from 0° to 30°, the keyhole region's depth within the molten pool exhibits measurements of 1.7656 mm, 1.5734 mm, and 1.5766 mm (see Fig. 9(a)). In contrast to the pool growth stage, the molten pool depth does not exhibit substantial growth during the stabilization stage. On the contrary, it displays a certain degree of retraction. Moreover, the molten pool depth remains relatively unaffected by changes in the lateral gravity angle. In the context of the keyhole, elevating the lateral gravity angle from 0° to 15° results in a diminishing keyhole depth, ranging from 1.5234 mm to 1.2812 mm, and at a lateral gravity angle of 30°, the keyhole depth further reduces to 1.2563 mm (see Fig. 9(b)). The overall pattern of alteration closely parallels that observed in molten pool depth. Similarly, during the pool growth stage, the keyhole at a lateral gravity angle of 30° exhibits a flatter profile, inclining in the direction of gravity, in contrast to the keyhole at 0°. This suggests that during the pool stabilization stage, varying lateral gravity angles indeed exert a discernible influence on both the molten pool depth and keyhole depth. However, in contrast to the pool growth stage, the pool depth and keyhole depth do not exhibit a continuous reduction with an increasing lateral gravity angle; rather, the keyhole depth and pool depth at 15° align closely with those observed at 30°, suggesting a nuanced influence during the pool stabilization stage.

The morphology of the keyhole captured through high-speed imaging using optical glass is illustrated in Fig. 10. Measurements reveal that during the keyhole growth phase, the depths at 0°, 15°, and 30° are 1.61 mm, 1.42 mm, and 1.11 mm, respectively. In the stable keyhole phase, the depths at 0°, 15°, and 30° are measured at 1.60 mm, 1.23 mm, and 1.20 mm, respectively, with a comprehensive error not exceeding 5%. This signifies the precision of the present simulation model. Observations of the keyhole morphology under experimental conditions indicate that with an increase in lateral gravity inclination, the noticeable trend of reduced keyhole depth becomes apparent. Specifically, during the keyhole growth phase, it exhibits a relatively unstable nature,

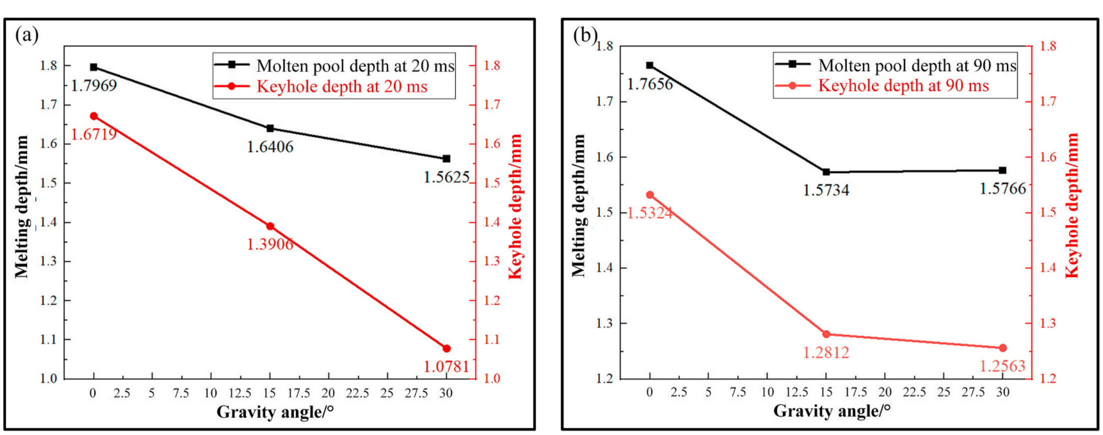

Fig. 9. The depth of the laser welding molten pool and the keyhole at different time intervals and under different gravitational angles: (a) t = 20 ms; (b) t = 90 ms.

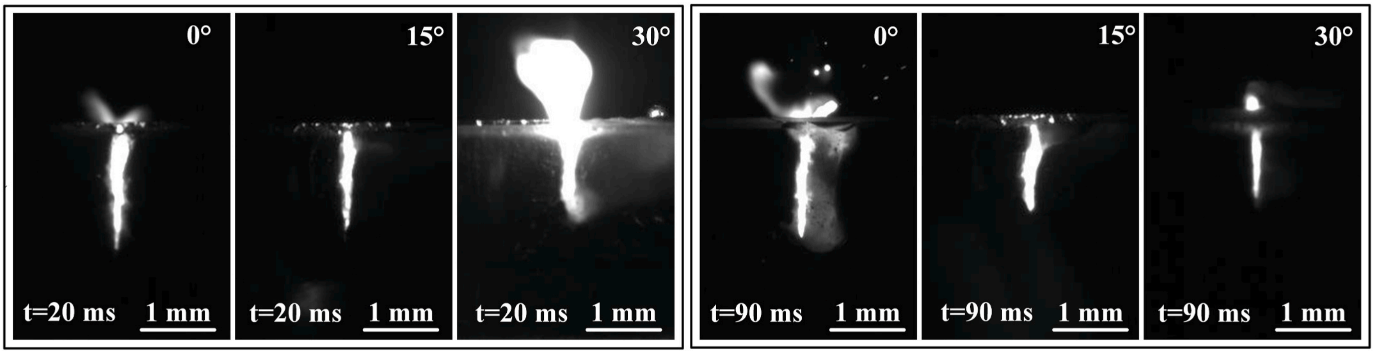

Fig. 10. The morphology of the keyhole at different time intervals and under different gravitational angles.

characterized by upper expansion, lower contraction, and significant undulations along its wall, rendering them less smooth. Conversely, in the stable keyhole phase, it displays a more uniform width, and the inner surface of the keyhole manifests higher smoothness.

4.3. Flow field characteristics

The study predominantly centered on scrutinizing the temperature field within the weld pool under varying lateral gravity angles, with a particular emphasis on elucidating the pronounced impact of lateral gravity on the flow dynamics within the weld pool. Fig. 11 illustrates the streamline distribution and velocity field within the cross-section during the growth stage under diverse lateral gravity angles. Fig. 11(a–c) depict the streamlines, whereas Fig. 11(d–f) illustrate the velocity fields. Upon scrutinizing the streamlines,it is evidently found that at a lateral gravity angle of 0°, the streamlines at the bottom of molten pool manifest a stratified flow pattern, demonstrating a homogeneous distribution. The upper region of the molten pool displays denser and more turbulent traces, while its uppermost section exhibits a more consistent stratified flow. When the lateral gravity angle is 15°, it can be observed that there is a pronounced layered flow at the bottom of the molten pool, while some areas in the middle of the molten pool exhibit more turbulent flow, located at the top side of the keyhole and in the middle of the molten pool. The upper section of the molten pool similarly exhibits a consistently stratified flow. In the case of a lateral gravity angle of 30°, the keyhole's flow pattern predominantly maintains a layered configuration.

Furthermore, the velocity fields elucidate that under a lateral gravity

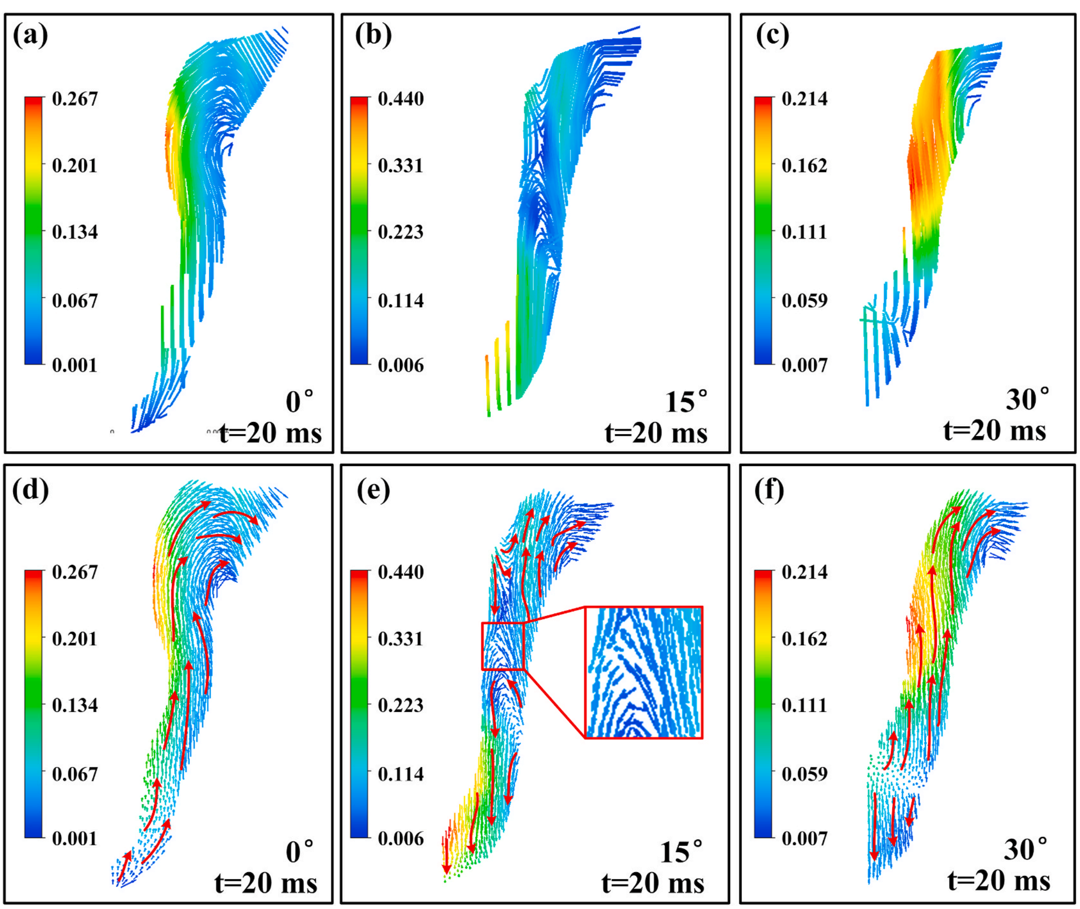

Fig. 11. Cross-sectional molten pool flow behavior in laser welding processes under different lateral gravity angle at 20 ms: (a, d) lateral gravity angle = 0°; (b, e) lateral gravity angle = 15°; (c, f) lateral gravity angle = 30°.

angle of 0°, the molten pool experiences a conspicuous Marangoni effect, influencing its overall flow. The fluid within the molten pool exhibits an upward flow pattern along the molten pool walls, with the highest flow velocity observed on the upper side of the keyhole wall. At a lateral gravity angle of 15°, the internal flow within the molten pool becomes turbulent, marked by the downward flow of fluid at the bottom of molten pool and the impingement of some fluid in the middle region against the keyhole wall. Towards the top of the keyhole, there is a partial downward flow, coupled with a convergence of more fluid towards the upper portion of the molten pool, giving rise to an overall state of disordered flow. Within the molten pool, notable convection phenomena are evident, with the region of highest flow velocity situated at the base of the keyhole. As the lateral gravity angle escalates to 30°, the prevalence of the Marangoni effect resurfaces, resulting in an upward flow within the upper region of the molten pool, while the lower section succumbs more to gravitational effects, inducing a downward flow. A well-defined demarcation delineates these two fluidic domains, with the apical zone of the keyhole wall hosting the utmost flow velocity. Pertaining to flow dynamics, during the expansion phase, at a lateral gravity angle of 15°, the internal fluid motion within the molten pool assumes a turbulent character, characterized by a pinnacle flow velocity reaching 0.440 m/s. At 0° and 30° lateral gravity angles, the intra molten pool flow exhibits a relatively stable demeanor, registering flow velocities of 0.267 m/s and 0.214 m/s, respectively, with an overall uniformity in the velocity distribution pattern.

Fig. 12 displays the streamlines and flow fields within the molten pool's cross-section during the stable stage (t = 90 ms) of the laser welding process under varying lateral gravity angles. In Fig. 12 (a, d), it is evident that at a lateral gravity angle of 0°, the distribution of molten pool streamlines deviates from that observed during the growth stage. The laminar flow at the bottom of the molten pool disappears, and the flow directions of the fluid become more varied Furthermore, a conspicuous vortex region manifests at the uppermost section of the molten pool. At a lateral gravity angle of 15°, the lower streamlines continue to exhibit a stratified flow pattern, although a notable vortex forms at the underside of the molten pool. The remainder of the region maintains characteristics similar to those observed during the growth stage. At a lateral gravity angle of 30°, a conspicuous vortex emerges in the upper region of the molten pool, while the distribution in other areas closely resembles that observed during the growth stage.

Furthermore, the velocity field analysis reveals that at a lateral gravity angle of 0°, the overall flow within the molten pool exhibits a marked Marangoni effect, accompanied by a minor vortex at the upper part. The area exhibiting the highest flow velocity is situated in the central section of the keyhole wall. At a lateral gravity angle of 15°, the degree of turbulence experiences a marginal reduction in comparison to the growth stage, signifying a comparatively more streamlined flow pattern. However, the overall flow trend remains largely unchanged, with the zone of maximum flow velocity still positioned at the base of the keyhole. At a lateral gravity angle of 30°, the fundamental flow pattern remains comparatively stable; however, the demarcation between the upper and lower fluid regions becomes more distinct, and the zone exhibiting the highest flow velocity relocates upwards. In contrast to the growth stage, the flow within the molten pool stabilizes during the

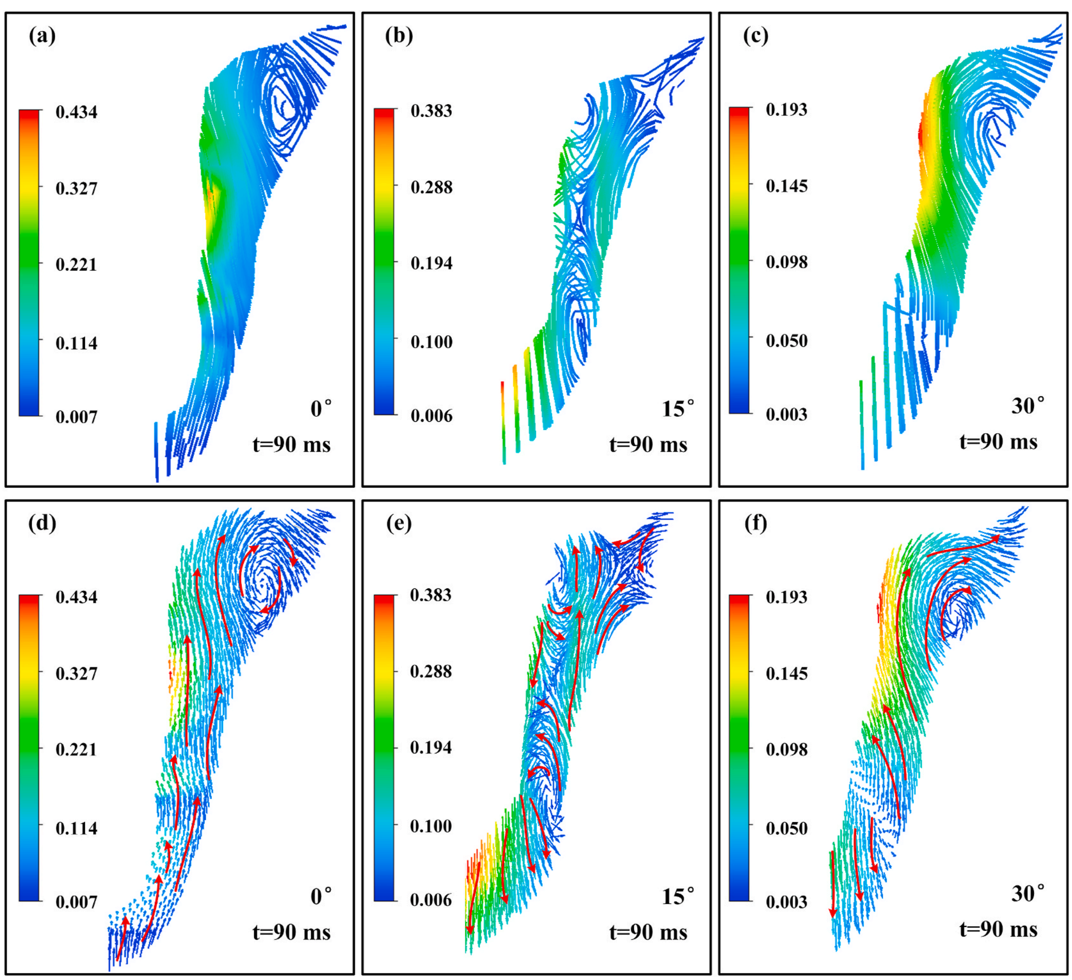

Fig. 12. Cross-sectional molten pool flow behavior in laser welding processes under different lateral gravity angle at 90 ms: (a, d) lateral gravity angle = 0°; (b, e) lateral gravity angle = 15°; (c, f) lateral gravity angle = 30°.

stable stage. Regarding flow velocity, it exhibits a gradual reduction with increasing lateral gravity angles. Specifically, as the lateral gravity angle transitions from 0° to 15°, the maximum flow velocity within the molten pool diminishes from 0.434 m/s to 0.383 m/s. Upon further elevation of the lateral gravity angle to 30°, the maximum flow velocity within the molten pool further diminishes to 0.193 m/s. Hence, it can be deduced that the lateral gravity angle exerts a substantial attenuating influence on the flow velocity within the molten pool.

Subsequently, the analysis narrows its focus to scrutinize the streamlines and flow patterns within the longitudinal cross-section of the molten pool during the growth stage (t = 20 ms) across diverse lateral gravity angles in the laser welding procedure. The specific simulation results are visually presented in Fig. 13. It is noticeable that at a lateral gravity angle of 0°, the fluid flow at the bottom of molten pool exhibits a relatively uniform, layered flow pattern. A pronounced vortex is discernible in the middle of the molten pool, while the flow in the surrounding regions appears relatively smooth. At a lateral gravity angle of 15°, the distribution of streamlines within the molten pool exhibits a uniform pattern without the presence of any discernible vortices, and the overall flow appears relatively consistent. However, at a lateral gravity angle of 30°, two pronounced vortices are observed within the molten pool. One vortex is located in the middle of the molten pool, while the other is induced by the presence of bubbles at the bottom of molten pool. The arrangement of streamlines within the middle of the molten pool exhibits a relatively high density, whereas the streamlines proximate to the bubbles at the bottom of molten pool display a relatively turbulent pattern.

Under a lateral gravity angle of 0°, during the growth stage at t = 20 ms, there is a phenomenon of reverse flow in the upper rear section of the molten pool, where a portion of the fluid moves in the direction of the keyhole side, leading to the formation of a unilateral bulge. On the lower side of the molten pool, the flow is directed towards the keyhole side; however, the presence of recoil pressure mitigates substantial bulging as a result of this flow pattern. The fluid on the front side of the keyhole ascends along the keyhole wall, and the region displaying the highest flow velocity is situated at the protrusion on the rear wall of the keyhole.

At a lateral gravity angle of 15°, the entire molten pool exhibits upward flow on the rear side, while the fluid on the front side of the molten pool undergoes a downward surge. The lateral walls of the molten pool exhibit a more consistent flow pattern with reduced impact from the recoil pressure, and the area with the highest flow velocity is situated on the lower side of the front wall of the keyhole. In the scenario where the lateral gravity angle is set at 30°, the keyhole exhibits restricted downward extension capacity and exerts a relatively lesser recoil pressure. Under the influence of gravity, it is susceptible to molten pool collapse, resulting in the emergence of bubbles within the molten pool during the growth stage. In the upper rear portion of the molten pool, a backflow phenomenon is observed, while the fluid on the keyhole wall side is influenced by the Marangoni effect, exhibiting an upward flow followed by a downward movement along the rear wall of the keyhole. A comparable circulation pattern is also evident in the vicinity of the pores, with the fluid exhibiting a circumferential flow around these voids. During this stage, the area with the highest flow velocity is situated in the central portion of the rear wall of the keyhole. Observations reveal that, during the growth stage, distinct lateral gravity angles primarily exert their influence on the keyhole's expansion dynamics. This influence is characterized by a reduction in the recoil pressure and an elevated likelihood of keyhole instability and subsequent collapse.

Following the stabilization of the molten pool (t = 90 ms), it becomes evident that the internal flow within the molten pool exhibits greater uniformity in contrast to the preceding growth stage. At a gravity angle of 0°, a conspicuous backflow phenomenon is observed at the uppermost section of the molten pool, signifying a flow direction counter to the prevailing overall flow. The flow at the bottom of the molten pool appears more chaotic. Nevertheless, the flow characteristics in other regions maintain a relatively uniform pattern, devoid of any conspicuous vortices. In the context of a 15° lateral gravity angle, the fluid dynamics within the molten pool's lower section manifest a stratified flow, with the maximum flow velocity localized at the bottom of molten pool, specifically at the rear portion and the upper extremity of the keyhole, as opposed to the keyhole's rear facet. Under a lateral gravity angle of 30°, the region exhibiting the most elevated flow rates extends more extensively to encompass the lower and central sectors of the molten pool.

Regarding the velocity field, the alterations in molten pool flow

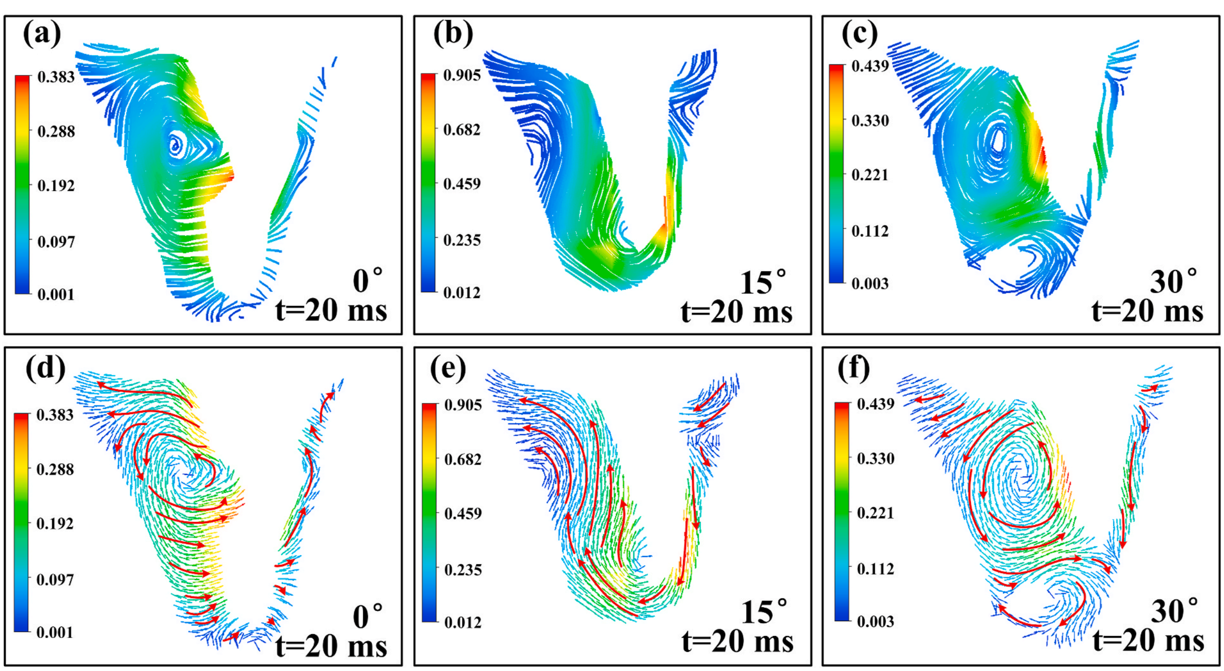

Fig. 13. Longitudinal sections molten pool flow behavior in laser welding processes under different lateral gravity angle at 20 ms: (a, d) lateral gravity angle = 0°; (b, e) lateral gravity angle = 15°; (c, f) lateral gravity angle = 30°.

patterns under various lateral gravity angles are comparatively less conspicuous when juxtaposed with the growth stage. In each scenario, notable backflow is discernible on the posterior aspect of the molten pool, concomitant with conspicuous upward flow at the posterior keyhole wall. Nevertheless, notable disparities are evident among various fluid regions. At a lateral gravity angle of 0°, an observable phenomenon occurs near the keyhole wall at the bottom of the molten pool, where fluid exhibits a tendency to flow towards the keyhole wall. At a lateral gravity angle of 15°, the region adjacent to the keyhole wall at the molten pool's bottom experiences a flow pattern directed towards the lower portion of the molten pool. With a lateral gravity angle of 30°, the molten pool's rear side continues to exhibit conspicuous backflow, accompanied by an upward flow along the rear wall of the keyhole. As the lateral gravity angle increases from 0° to 15°, a noticeable reduction in flow velocity is observed, with fluid velocity decreasing from 0.781 m/s to 0.569 m/s. At a lateral gravity angle of 30°, the fluid velocity diminishes to 0.411 m/s. This suggests that augmenting the lateral gravity angle significantly attenuates the fluid's flow velocity within the molten pool and alters the flow pattern at its base. Concerning the keyhole, elevating the lateral gravity angle results in a diminished keyhole depth, imparting a heightened smoothness to the keyhole overall, with a propensity for the keyhole wall to align with the direction of gravity.

Fig. 14 illustrates the streamlines and the distribution of velocity fields on the upper surface of the molten pool during the growth stage, at various lateral gravity angles in laser welding. Examining the streamlines distribution, it becomes evident that at a lateral gravity angle of 0°, the streamlines on the upper surface at the molten pool's rear exhibit a relative uniformity, contrasting with an area of heightened flow turbulence on the front side of the molten pool. At a lateral gravity angle of 15°, the streamlines distribution loses its uniformity, and the high-velocity region within the molten pool draws nearer to the keyhole. The distribution of streamlines on the front side of the molten pool assumes a heightened level of chaos. When the lateral gravity angle is 30°, the flow condition on the upper surface at the rear of the molten pool takes on a notably more disorderly disposition. Nevertheless, the distribution of fluid streamlines on the gravity side and the front side of the molten pool maintains a relatively uniform configuration.

Regarding the velocity field, discernible variations are evident in the flow dynamics on the upper surface of the molten pool. At a lateral gravity angle of 0°, the flow field on the molten pool's surface manifests a homogeneous distribution, characterized by the radial flow of fluid from the central region towards the periphery. Additionally, fluid exhibits an upward flow from the lower side of the molten pool's frontal region. This phenomenon is consistent in the same position when the lateral gravity angle is set at 15°. Furthermore, at lateral gravity angles of 15° or 30°, the fluid on the molten pool's surface demonstrates a tendency to flow in the direction of gravity.

Within the context of Fig. 15, the visualization of streamlines and the velocity field distribution on the upper surface of the molten pool elucidates the influence exerted by varying lateral gravity angles during the laser welding's stable stage. The insights drawn from Fig. 15 underscore the relatively modest impact of varying lateral gravity angles on the comprehensive streamlines distribution at the rear section of the molten pool. A consistent flow pattern is discernible, marked by the fluid's motion from the side of the keyhole toward the rear section of the molten pool. On the front side of the molten pool, a discernible influence can be discerned; nevertheless, in the grander scheme of things, its impact remains comparatively modest in contrast to the growth stage. At a lateral gravity angle of 15°, an observable vortex manifests on the anterior side of the molten pool, whereas, at a lateral gravity angle of 30°, significant alterations become evident in the configuration of the keyhole.

In the analysis of the velocity field during the stable stage of the molten pool, it becomes apparent that the flow characteristics on the upper surface of the molten pool exhibit minimal fluctuations, whereas some alterations in the configuration of the keyhole opening are discernible (see Fig. 16). With the augmentation of the lateral gravity angle, there is a concurrent expansion in the upper aperture region of the keyhole. This phenomenon can be attributed to the unique downward growth dynamics of the keyhole, which is a consequence of varying lateral gravity angles, consequently causing an augmentation in the keyhole's width both laterally and in the front-back orientation.

4.4. Analysis of keyhole forces and molten pool flow dynamics

To elucidate the dynamic alterations in the flow patterns within the molten pool stemming from the metamorphosis of the keyhole's morphology under varying gravitational angles, a mechanical analysis of the stress conditions within the keyhole wall region is undertaken. This analysis is graphically depicted in Fig. 17. The forces acting on the keyhole wall region primarily include recoil pressure (P_v), surface tension (F_s), fluid hydrostatic pressure (P_m), fluid dynamic pressure (F_v), vapor-metal friction force (F_f) and gravity(g).

In the laser welding process of Ti–6Al–4V titanium alloy under diverse gravitational influences, it becomes evident that gravity exerts a substantial influence on the configuration and evolution of the keyhole. Fig. 17(a–c) intricately illustrates the evolving features of keyhole morphology as the gravitational angle undergoes variation. This dynamic transition is marked by a progressive reduction in the keyhole's

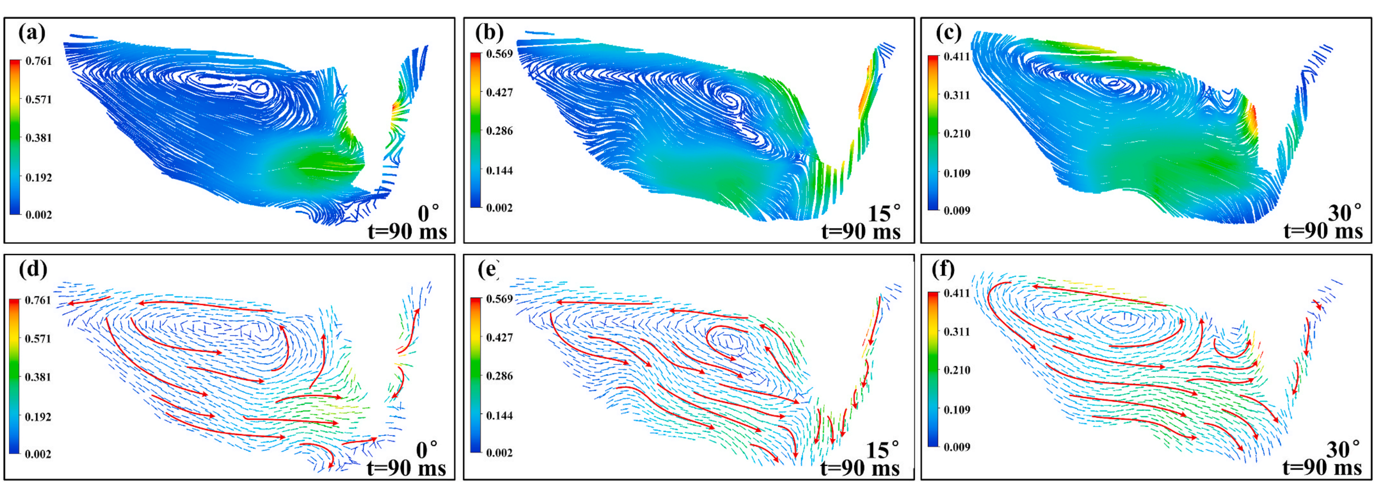

Fig. 14. Longitudinal sections molten pool flow behavior in laser welding processes under different lateral gravity angle at 90 ms: (a, d) lateral gravity angle = 0°; (b, e) lateral gravity angle = 15°; (c, f) lateral gravity angle = 30°.

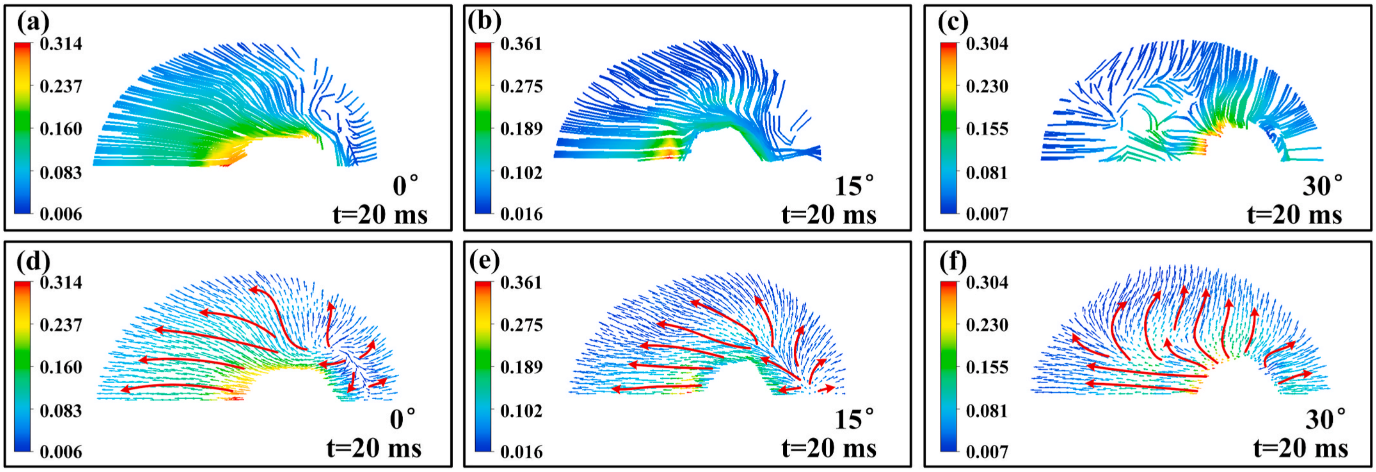

Fig. 15. Upper surface sections molten pool flow behavior in laser welding processes under different lateral gravity angle at 20 ms: (a, d) lateral gravity angle = 0°; (b, e) lateral gravity angle = 15°; (c, f) lateral gravity angle = 30°.

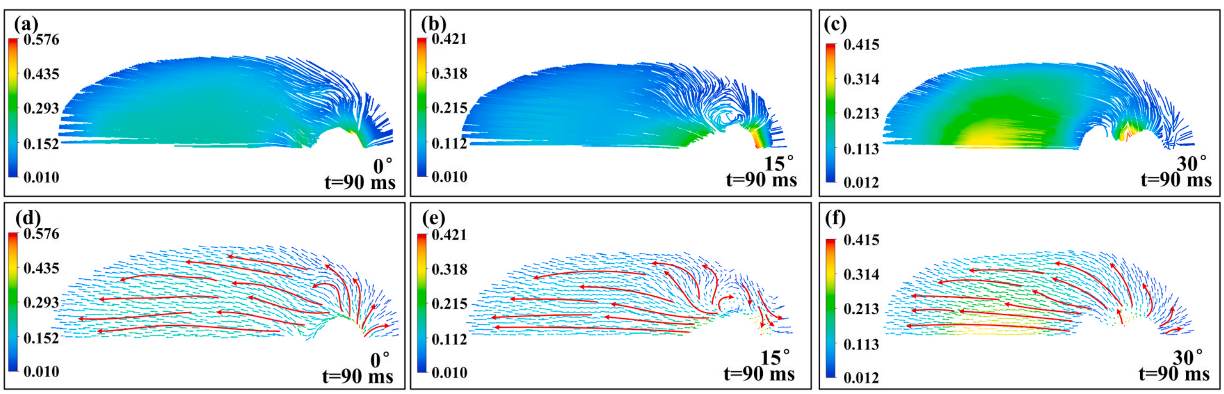

Fig. 16. Upper surface sections molten pool flow behavior in laser welding processes under different lateral gravity angle at 90 ms: (a, d) lateral gravity angle = 0°; (b, e) lateral gravity angle = 15°; (c, f) lateral gravity angle = 30°.

depth, juxtaposed with a concomitant widening of its aperture. First, the stress conditions at points A, B, and C at the top of the keyhole are analyzed and described as follows:

Point A:

$$P_{mx} < F_{sx} + F_{fx} + F_{vx} + P_{vx} \tag{14}$$

$$P_{my} + F_{vy} + F_{sy} + F_{fy} < g + P_{vy} \tag{15}$$

Point B:

$$P_{mx} + F_{vx} < F_{sx} + F_{fx} + g_x + P_{vx} \tag{16}$$

$$P_{my} + F_{vy} + F_{sy} + F_{fy} < g_y + P_{vy} \tag{17}$$

Point C:

$$P_{mx} < F_{vx} + F_{sx} + F_{fx} + g_x + P_{vx} \tag{18}$$

$$P_{my} + F_{vy} + F_{sy} + F_{fy} > g_y + P_{vy} \tag{19}$$

where P_{mx}, F_{vx}, F_{sx}, F_{fx}, g_x, P_{vx}, P_{my}, F_{vy}, F_{sy}, F_{fy}, g_y and P_{vy} are the X-axis, Y-axis, and Z-axis components of P_m, F_v, F_s, F_f, g and P_v respectively. Through an exhaustive examination of the force conditions at the three aforementioned points, it is discerned that the resultant force along the X-axis persists in signifying the keyhole's continued expansion outward. The alteration in the gravitational orientation amplifies the force acting in that specific direction, facilitating a more seamless and unhindered expansion of the keyhole's upper region. Consequently, with the elevation of the lateral gravity angle, there is a concomitant augmentation in the width of the keyhole's upper aperture. Conversely, the resultant force in the Y-axis direction signifies an inclination for the upper section of the keyhole to ascend. With the elevation of the lateral gravity angle, the constraint on the force in that specific direction gradually diminishes. Consequently, as the lateral gravity angle increases, the uppermost part of the keyhole gradually assumes a more flattened profile.

Point D:

$$F_{sx} + F_{fx} + P_{mx} + F_{vx} < P_{vx} \tag{20}$$

$$P_{vy} + F_{vy} + F_{sy} + F_{fy} < g_y + P_{my} \tag{21}$$

Point E:

$$F_{sx} + F_{fx} + P_{mx} + F_{vx} < P_{vx} + g_x \tag{22}$$

$$P_{my} + F_{sy} + F_{fy} < g_y + P_{vy} + F_{vy} \tag{23}$$

Point F:

$$P_{mx} + F_{vx} < P_{vx} + F_{sx} + F_{fx} + g_x \tag{24}$$

$$P_{my} + F_{vy} + F_{sy} + F_{fy} < g_y + P_{vy} \tag{25}$$

In the scrutiny of the forces acting upon points D, E, and F along the keyhole wall, it is discernible that the force in the X-axis direction undergoes continuous fluctuations but predominantly maintains a state of

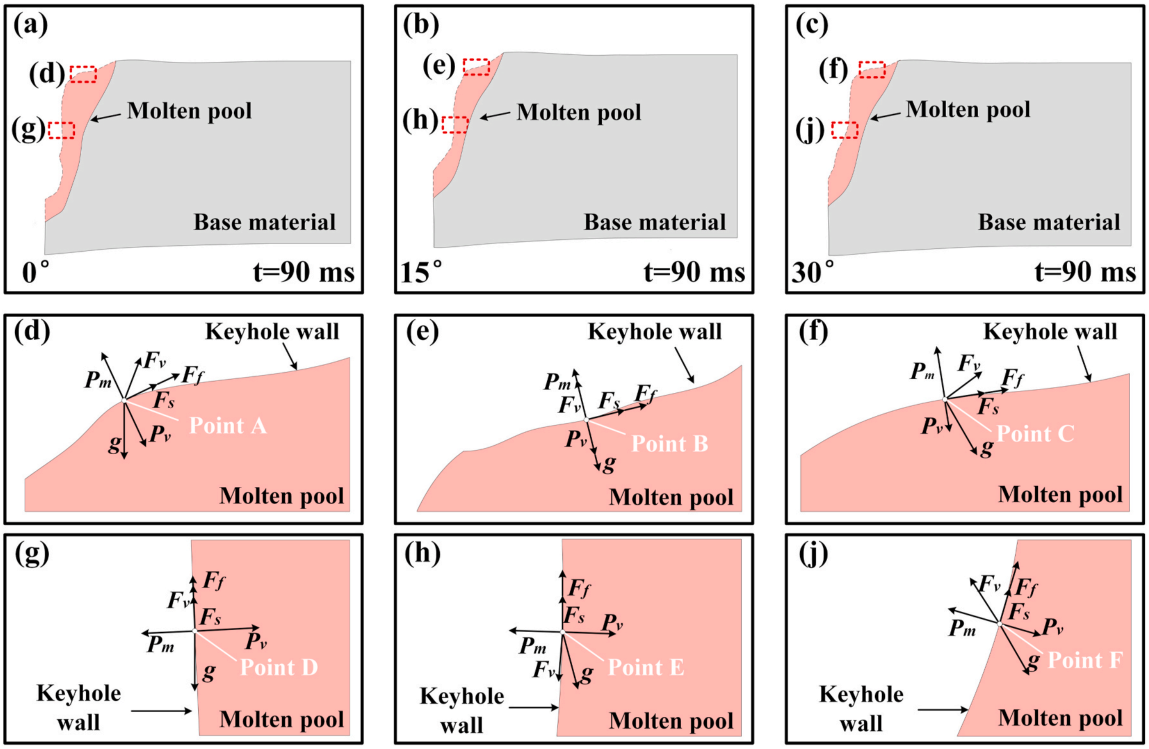

Fig. 17. Evolution of local stress conditions in the keyhole wall region under different gravitational angles: (a–c) Process of keyhole evolution under different gravitational effects; (d–f) Stress conditions at the top of the keyhole (g–j) Stress conditions in the keyhole wall.

equilibrium. The resultant force is directed towards the inner surface of the keyhole wall, applying compressive forces that counteract wall closure. Regarding the force along the Y-axis direction, it experiences notable oscillations and fluctuations. At a lateral gravity angle of 0°, the molten pool wall assumes a vertical and unblemished profile, while the internal fluid exhibits an upward flow. The resultant force in the Y-axis direction similarly aligns with an upward trajectory. Nonetheless, upon introducing a lateral gravity angle of 15°, the flow dynamics within the molten pool shift towards a downward direction, leading to a corresponding alteration in the resultant force's direction, which also descends accordingly. As the lateral gravity angle escalates to 30°, the resultant force in the Y-axis direction at the designated point reverts to an upward vector, instigating an upward flow of the lower fluid that exerts pressure on the keyhole. Consequently, the keyhole wall no longer maintains a vertical orientation, adopting an arched configuration at a specific angle.

Variations in the lateral gravity angle induce changes in the morphology of the keyhole wall, subsequently impacting the fluid dynamics within the molten pool (see in Fig. 18). During the stable phase of the molten pool, at a lateral gravity angle of 0°, the internal fluid

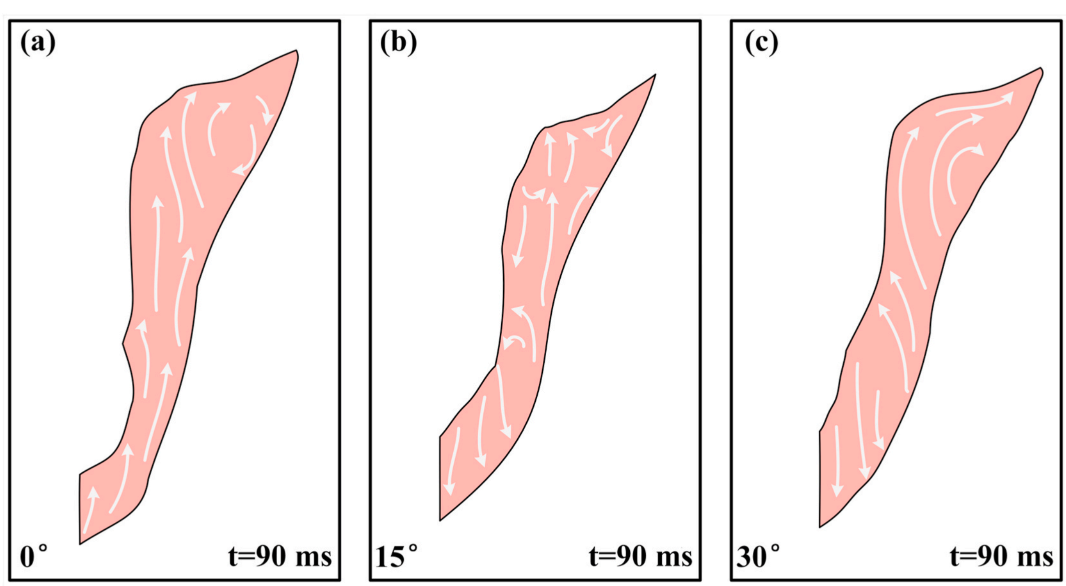

Fig. 18. The molten pool flow during laser welding under different lateral gravity angles: (a) lateral gravity angle = 0°; (b) lateral gravity angle = 15°; (c) lateral gravity angle = 30°.

dynamics maintain pronounced consistency, primarily governed by the Marangoni effect, resulting in upward flow. However, at a lateral gravity angle of 15°, gravitational forces acting on the keyhole wall influence the flow behavior of the molten pool fluid, causing fluids in different regions to flow in diverse directions. Additionally, gravity influences the recoil pressure within the molten pool, affecting the internal fluid dynamics of the keyhole and leading to turbulent states. Under a lateral gravity angle of 30°, forces applied to the keyhole wall reach a state of relative equilibrium, resulting in the division of the internal keyhole flow into two distinct regions. In these regions, fluid motion is influenced by both the Marangoni effect and gravity, causing variations in flow direction.

5. Conclusions

This study delves into an examination of the impact of lateral gravity angles on the temperature field, streamlines, and velocity field within the context of laser welding, employing flow field simulations. The research outcomes can be summarized as follows.

(1). The depth of both the molten pool and the keyhole exhibits a consistent reduction as the lateral gravity angle increases. Nevertheless, the magnitude of this influence exhibits variations at distinct phases of the process. The findings elucidate that alterations in the lateral gravity angle induce an inclination of the keyhole in the direction of gravity, thereby curtailing its downward expansion, along with that of the molten pool.

(2). The developmental trajectory of the molten pool is notably responsive to alterations in the lateral gravity angle, resulting in a perturbation of the fluid flow dynamics within the molten pool. In response to variations in the lateral gravity angle, the initially ascending fluid driven by the Marangoni effect undergoes a transition to a downward trajectory, converging towards the molten pool's lower regions, while the upper portion of the molten pool maintains its upward flow.

(3). During the stable phase of the molten pool, the fluid dynamics on both sides of the keyhole exhibit a relatively constant state. In response to variations in force conditions, the lateral gravity angle incrementally diminishes the flow velocity of the molten pool fluid. When the lateral gravity angle reaches 30°, the maximum flow velocity within the molten pool markedly decreases by 55.6%.

(4). The influence of gravity on the molten pool is primarily manifested in the mechanical forces exerted on the fluid. Changes in force characteristics under different gravity angles are predominantly observed in the interplay between gravity and the recoil pressure, where an increase in the lateral gravity angle results in a corresponding reduction in recoil pressure. This reduction subsequently restrains the expansion of the keyhole, making it more prone to collapse and inducing a tilt of the keyhole along the gravitational direction.

(5). Throughout the entire molten pool growth phase, the impact of gravity induces significant alterations in the flow patterns of the fluid located on the upper surface of the molten pool, thereby enhancing its movement in the direction aligned with gravity.

CRediT authorship contribution statement

Chao Ma: Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yue Li:** Writing – review & editing, Supervision, Resources, Project administration, Investigation. **Lihong Cheng:** Writing – review & editing, Software, Investigation, Formal analysis. **Yanqiu Zhao:** Writing – review & editing. **Jianfeng Wang:** Writing – review & editing, Supervision, Conceptualization. **Xiaohong Zhan:** Writing – review & editing, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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